



University of Asia Pacific

Department of Computer Science and Engineering

CSE 450: VLSI Lab

LAB REPORT

Experiment Number: 01

Experiment Title: Chip-Design (Full Adder-8bits)

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1. Title of the Experiment

Design and Simulation of a 8-Bit Full Adder

2. Objectives

This experiment's primary goal is to use Verilog HDL to develop and simulate an 8-bit Full Adder.

The particular goals are:

- to comprehend how an 8-bit binary adder works.
- to use eight 1-bit Full Adders to create an 8-bit Full Adder.
- to comprehend the spread of carry between various stages.
- to use Xilinx ISE to implement the design.
- to use ISim Simulator to simulate the circuit.
- to confirm the circuit's ultimate carry-out and sum.

3. Software and Tools Used

Hardware Needs

- Computer/Laptop
- At least 4 GB of RAM
- Mouse and Keyboard

Software Requirements

- Verilog HDL Simulator Xilinx ISE 14.7
- VirtualBox (if a virtual computer is used to run Xilinx ISE)
- Verilog HDL was used to create the Full Adder, and Xilinx ISE's ISim Simulator was used to simulate it.

4. Design Methodology

Eight 1-bit Full Adders are cascaded together to create the 8-bit Full Adder.

Two input bits and one carry input are received by each Full Adder. One Full Adder's carry output is linked to the subsequent Full Adder's carry input.

The fundamental framework is:

FA0 → FA1 → FA2 → FA3 → FA4 → FA5 → FA6 → FA7

The external Cin is received by the first Full Adder, and the final Cout is generated by the last Full Adder.

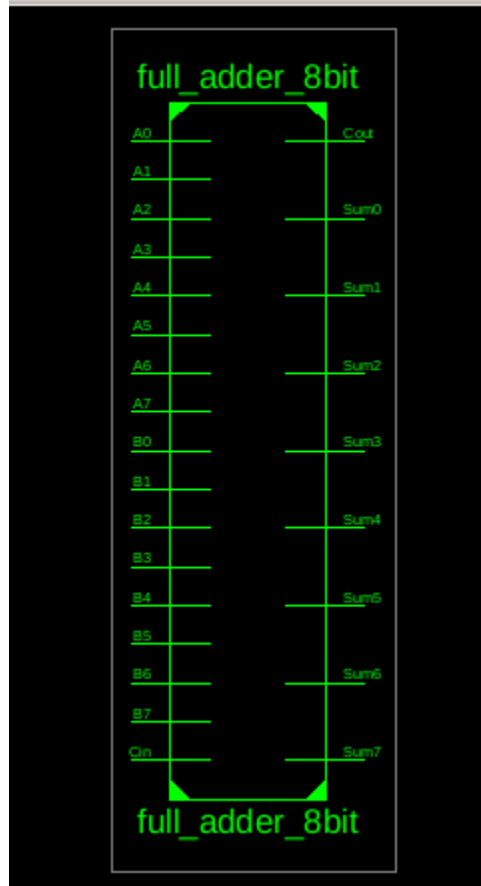


Fig: RTL schematic diagram

6. Output Analysis / Observations

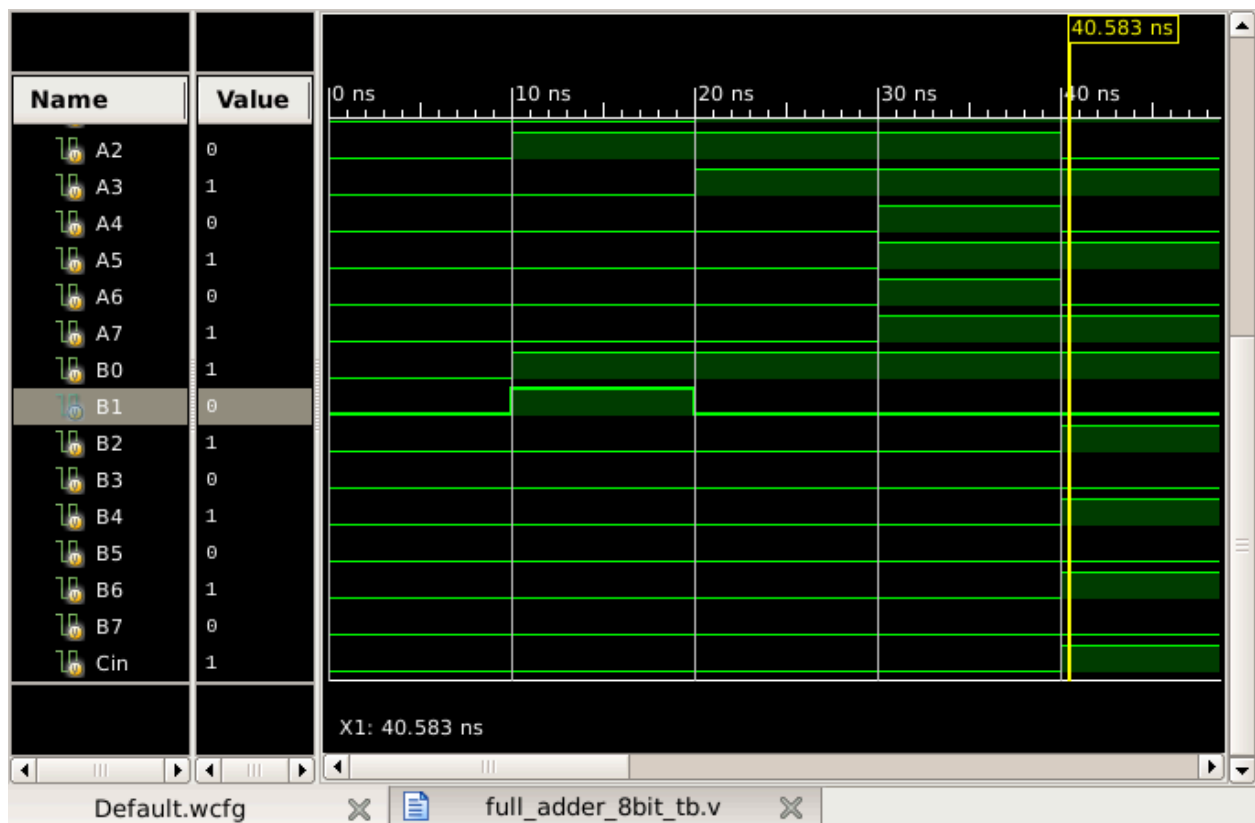
The 8-bit Full Adder was simulated using ISim. Different binary values were applied to the inputs A, B, and Cin.

Some of the important test cases are shown below:

A	B	Cin	Expected Output
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00000000	00000000	0	00000000
00000101	00000011	0	00001000
00001111	00000001	0	00010000
11111111	00000001	0	Sum = 00000000, Cout = 1
10101010	01010101	1	Sum = 00000000, Cout = 1

The ISim waveform showed that the Sum and Cout outputs changed correctly according to the input values.



The observed simulation output matched the expected truth table.

7. Result

The 1-bit Full Adder was successfully designed using Verilog HDL, and it was modeled using Xilinx ISE and ISim. The simulation produced the correct Sum and Carry-out for each possible combination of inputs.

Consequently, the experiment's objective was successfully achieved.

8. Github Repository Commits

GitHub Repository:

<https://github.com/Akifhossain29/ISE-14.7-VLSI>

GitHub Commit Link:

<https://github.com/Akifhossain29/ISE-14.7-VLSI/commits/main/>

The repository contains the Verilog source code and related files used for this experiment.

9. Conclusion

In this experiment, I learned how to design an 8-bit Full Adder by connecting eight 1-bit Full Adders. I also learned how carry is passed from one stage to the next stage.

The design was implemented using Verilog HDL and simulated using Xilinx ISE and ISim. The simulation results were correct for the tested binary inputs. Therefore, the experiment was successfully completed and the objectives were achieved.